# United International University
## Department of Computer Science and Engineering

CSE 3421: Software Engineering  Final: Spring 2026

Total Marks: 40  Time: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

---

*Answer all five questions. The numbers on the right of the questions denote their marks.*

1. Design **UML diagrams** using the **appropriate design pattern** for the following scenarios: [CO2]

   (a) IntelliWrite is an AI-powered writing assistant used for drafting emails, reports, and social media posts. The application supports multiple text generation modes, including Professional Writing, Creative Writing, and Technical Writing. Users can switch between these generation modes at runtime depending on their needs. The system should allow additional writing modes to be added in the future without modifying existing code. (4)

   (b) IntelliWrite interacts with language models via the standard LLMService interface for text generation. The system was originally designed to work with OpenAI's GPT service. The team now wants to integrate Anthropic's Claude as an additional AI provider. Since Claude uses a different request and response format, the team must ensure that the new providers' format works seamlessly with the existing application without modifying IntelliWrite's internal code. (4)

2. (a) Assume the following scenario, where a student is using the university course registration system in order to successfully complete his registration for the next semester. While using the system, he noticed that the system uses the same terminology, button names, icons and page layouts throughout the different pages. When he needed to quickly search for his desired course and faculty information, he was able to use a keyboard shortcut. Once the course was successfully registered, the system displayed a confirmation message with the course details. However, when he tried to register for Data Structures and Algorithms II without actually completing Data Structure and Algorithms I first, the system blocked him from proceeding, and a pop-up message explained the reason behind it. He later realized that the pre-requisite courses were displayed on the first page of the system only, but by the time he reached the actual registration page, he had long forgotten all that information. Finally, when he was almost done with his work, he accidentally dropped an important course from his list, but the system provided no ways to fix the action. (5)
   **Explain** which of **Schneiderman's** Golden Rules are **maintained** within the aforementioned scenario, and which rules are **violated**. [CO1]

   (b) **Explain** the **concepts** on which the effectiveness of **usability of UI** depends on. [CO1] (3)

3. (a) A software company is developing an online food delivery platform. The system will consist of independent facilities such as User Management, Restaurant Management, Order Processing, Payment Processing, Delivery Tracking, and Notifications. Each facility will be developed by separate teams and may need to be updated or deployed independently without affecting the others. The platform is expected to serve millions of users daily. (5)
   Which architectural **pattern** will be **best suited** for the software above? **Explain** with **proper reasons**.
   If the company later decides to simplify the platform for a small local restaurant chain with only a few thousand users and a single development team, which architectural **pattern** should be **adopted** then and **why**? [CO2]

   (b) **Explain** the **similarities** and **differences** between **Acceptance** testing and **Beta** testing. [CO2] (3)

4. (a) A software development team at Undefeatable Company is building a healthcare application, yet they continue to use outdated patient data despite knowing it may compromise accuracy. In one instance, a developer secretly accepts additional payment from a stakeholder to approve a substandard feature. Furthermore, employees involved in the project are compensated unfairly relative to their workload. On another occasion, a developer reviews a peer's work without obtaining their consent. Additionally, some team members make no effort to stay updated with current policies and legal regulations relevant to the company. (5)
Which of the ACM **principles** of ethics are **violated** in the above scenario? **Explain** with proper reasoning. [CO3]

(b) How are modern software systems changing from traditional code-centric development to **cloud-native**, **API-based** and **continuously delivered** products? Briefly **describe**. [CO2] (3)

5. (a) Find the **Earliest Start, Earliest Finish, Latest Start, Latest Finish, Float and Critical Path** using the **Critical Path Method** for the following tasks: [CO2] (6)

| Tasks | Duration (Weeks) | Precedents |
|---|---|---|
| A | 3 | - |
| B | 2 | - |
| C | 4 | - |
| D | 5 | A |
| E | 3 | B |
| F | 6 | C |
| G | 4 | D,E |
| H | 2 | D |
| I | 5 | F,G |
| J | 3 | H |
| K | 4 | I,J |

(b) A company assigned a **deadline** of **18 weeks** for the above project to your team. Can your team **complete** the project in time? **Justify** your answer. (2)